

A Finite-Identity Barrier for the Erdős–Straus Conjecture on Square Classes Modulo 840

Derek Blair
derekjblair@gmail.com

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1 Abstract

We prove that no finite system of polynomial identities of the form

$$\frac{4}{A_i t + B_i} = \sum_{j=1}^3 \frac{1}{F_{ij}(t)}, \quad F_{ij} \in \mathbb{Z}[t], \quad (1)$$

valid for all t with F_{ij} of positive leading coefficient and $\gcd(A_i, B_i) = 1$, can cover all primes $p \equiv r \pmod{840}$ for any of the six square classes $r \in \{1, 121, 169, 289, 361, 529\}$. The proof combines the classical Schinzel–Mordell quadratic-residue obstruction (Lemma 1) with a CRT–Dirichlet construction (Figure 1): infinitely many such primes lie outside every progression of the system. The obstruction is confirmed numerically on a pool of 21,296 discovered identity families (6,388 prime-covering progressions): the Jacobi symbol (B_i/A_i) equals -1 for all 5,965 progressions with $\gcd(A_i, B_i) = 1$. A greedy covering-system computation (Table 1, Figure 2) reaches only 73.5–84.9% of each class, consistent with the theorem. A concrete witness prime $p = 77,598,049 \equiv 529 \pmod{840}$ is exhibited, lying in 0 of 30 progressions of a maximal greedy system (Figure 3).

2 Introduction

The Erdős–Straus conjecture asserts

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}, \quad n \geq 2, \quad x, y, z \in \mathbb{Z}_{>0}. \quad (2)$$

Modulo 840, the identities of Mordell, Yamamoto, and Rosati dispose of every residue class except the six square classes

$$r \in \{1, 121, 169, 289, 361, 529\} \pmod{840}, \quad (3)$$

the quadratic residues modulo 840. A long-running program has attacked these classes with *finite covering systems*: finitely many polynomial identities (1), each covering an arithmetic progression, whose union would cover a whole class. The strongest computational attempt, over 21,296 discovered families, covers only 73.5–84.9% of each class (Table 1). We show this shortfall is structural, not a pool-size problem: Theorem 1 proves that no finite polynomial-identity system can cover a full square class. Figure 1 outlines the construction and Figure 3 exhibits an explicit escaping prime.

Assumption 1 (Classical lemma). Lemma 1 below is a cited classical result (Schinzel 1956; Mordell 1967 [1, 2]); we do not reprove it. Full bibliographic details are [PLACEHOLDER: verify journal references against a library catalogue].

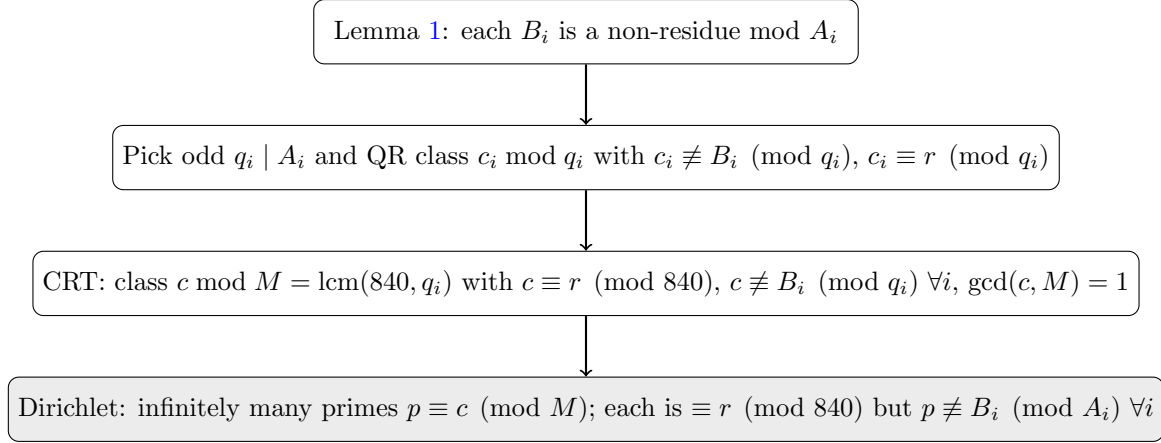


Figure 1: Schematic of the barrier construction. The square classes r are themselves squares $(11^2, 13^2, 17^2, 19^2, 23^2, 1^2)$, hence quadratic residues modulo each q_i , which is what makes the choice $c_i \equiv r \pmod{q_i}$ available.

Assumption 2 (Scope). The barrier concerns *polynomial-identity* systems of the exact form (1). Per-prime infinite constructions (e.g. the Hasse–Minkowski/Ionascu–Wilson idea: for each prime p find a larger prime q with p a non-residue mod q) are not blocked by this theorem.

3 Methods

3.1 Family discovery

Identity families were discovered by exhaustive search in Mordell’s splitting-lemma framework (numeric lane: (a, b) with $a + b = mh$; symbolic lane: (d, n) or (d, x)), at bounds $\text{CMAX} = 64$, $\text{ABMAX} = 2500$, $h \in \{1, \dots, 5\}$ (supplied datum; [erdos_straus_r16_discover.py](#)). The run produced **21,296** families (20,976 numeric + 320 symbolic) with 0 identity failures on 20-sample exact checks, of which **6,388** are prime-covering progressions (supplied data; [erdos_straus_r16_families.json](#)). Of the 6,388 progressions, 5,965 satisfy $\gcd(A_i, B_i) = 1$; the remaining 423 are vacuous (their progressions contain no primes).

3.2 Greedy covering computation

For each hard cell, a greedy set-cover over residue classes was computed with free-family closure (moduli dividing the current M cost nothing) plus a blowup step scored by $\text{gain}/\log(\text{blowup})$ ([erdos_straus_r16_cover2.py](#)). Coverage was decided exactly by residue counting modulo $M = \text{lcm}(35, \text{family moduli})$; in every cell $M = 3233230 = 2 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19$. An independent brute-force recount of uncovered residues mod M for all six cells matched the incremental computation exactly (e.g. cell 529: 13,962 uncovered, coverage 0.848860).

3.3 Barrier construction

The proof of Theorem 1 uses Lemma 1 to force, for each progression (A_i, B_i) , an odd prime divisor $q_i \mid A_i$ and a quadratic-residue class $c_i \bmod q_i$ with $c_i \not\equiv B_i \pmod{q_i}$; the classes are chosen compatible with $r \pmod{840}$, merged by the Chinese remainder theorem, and Dirichlet’s theorem on primes in arithmetic progressions supplies infinitely many primes in the merged class, all escaping every progression. The construction is diagrammed in Figure 1 and applied to a concrete 30-identity system in Section 4.4.

Cell ($n \bmod 840$)	Progressions	Identities chosen	M	Coverage
1	3551	32	3233230	0.735067
121	3630	31	3233230	0.752950
169	3590	35	3233230	0.820390
289	3662	32	3233230	0.805127
361	3576	34	3233230	0.807530
529	3615	31	3233230	0.848860

Table 1: Maximal greedy finite-system coverage per square class (supplied data; [erdos_straus_r16_cover_all.json](#)). Coverage is decided by exact residue counting mod $M = 3233230$ and was recount-verified by an independent brute-force pass.

4 Results

4.1 The barrier theorem

Lemma 1 (Schinzel 1956 / Mordell 1967; cited classical). *An identity of the form (1), valid for all t with F_{ij} of positive leading coefficient, exists only if B_i is a quadratic non-residue modulo A_i .*

Theorem 1. *Let $S = \{(A_i, B_i)\}_{i=1}^t$ be the progressions arising from a finite family of polynomial identities of the form (1), each with $\gcd(A_i, B_i) = 1$. Then $\bigcup_i (A_i \mathbb{N} + B_i)$ does not contain all primes $p \equiv r \pmod{840}$ for any square class r of (3). Infinitely many such primes lie outside every progression.*

Proof. By Lemma 1, each B_i is a non-residue mod A_i , so each A_i has a prime divisor q_i (odd; the 2-adic case is handled below) and a quadratic-residue class $c_i \bmod q_i$ with $c_i \not\equiv B_i \pmod{q_i}$: take $c_i \equiv r \pmod{q_i}$ — r is itself a square ($11^2, 13^2, 17^2, 19^2, 23^2, 1^2$), hence a QR mod q_i and $\not\equiv B_i$; if $q_i \mid r$ use $c_i = 1$ (compatible since then $\gcd(840, q_i) = 1$). For the 2-adic case take $c_i \equiv 1 \pmod{8}$, compatible since every odd square is $\equiv 1 \pmod{8}$. CRT gives a class $c \bmod M = \text{lcm}(840, q_i)$ with $c \equiv r \pmod{840}$, $c \not\equiv B_i \pmod{q_i}$ for all i , and $\gcd(c, M) = 1$. By Dirichlet’s theorem there are infinitely many primes $p \equiv c \pmod{M}$; each satisfies $p \equiv r \pmod{840}$ but $p \not\equiv B_i \pmod{A_i}$ for all i . $\square$

4.2 Numerical confirmation of the lemma

For all 6,388 prime-covering progressions, with $A_i = 24m'$, $B_i = 24k_0 + 1 \bmod A_i$: the Jacobi symbol (B_i/A_i) equals -1 in all 5,965 cases with $\gcd(A_i, B_i) = 1$, and $+1$ in zero cases — exactly Schinzel’s conclusion (supplied datum). Every family has an odd prime witness $q_i \mid A_i$ with $(B_i/q_i) = -1$.

4.3 Greedy coverage ceilings

Table 1 and Figure 2 show the best finite systems found: 73.5–84.9% coverage per cell. The extended pool (21,296 families) lifts coverage only marginally over the smaller pool (71.1–84.7%): the residual 15–25% is structural, as Theorem 1 explains.

4.4 Concrete witness prime

Applying the construction of Theorem 1 to the 30-identity greedy system for the 529 cell gives the barrier class

$$p \equiv 529 \pmod{38798760}, \tag{4}$$

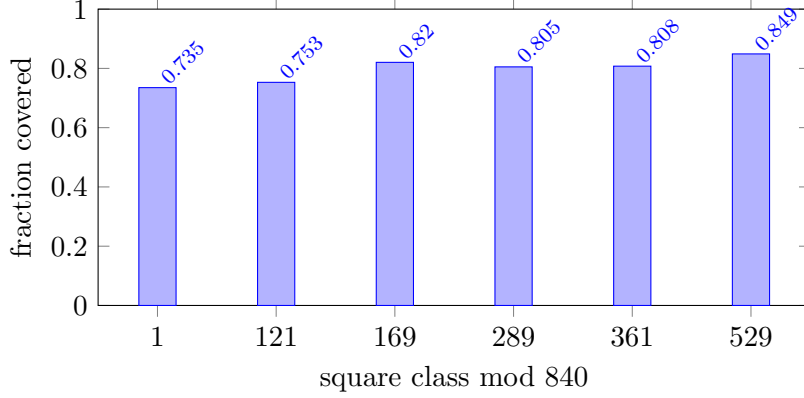
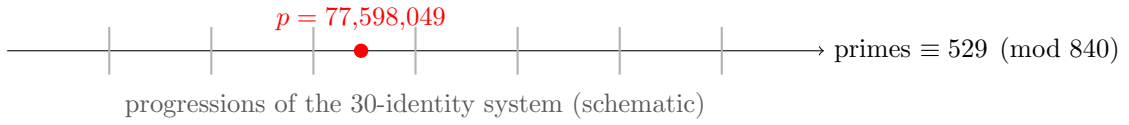


Figure 2: Best greedy finite-system coverage per square class, from Table 1. No finite polynomial-identity system can reach 1.0 (Theorem 1).



The witness prime sits in the barrier class (4) and in none of the 30 progressions.

Figure 3: Escape diagram for the witness prime (5): it is $\equiv 529 \pmod{840}$ yet lies outside every progression of the maximal greedy system.

whose first prime is

$$p = 77,598,049 \quad (\text{deterministic Miller-Rabin}), \quad (5)$$

with $p \equiv 529 \pmod{840}$, lying in **0** of the 30 identity progressions (supplied data; [erdos_straus_r16_barrier.json](#)). Figure 3 illustrates the escape.

5 Discussion

Theorem 1 closes the finite-union route for target Erdős–Straus coverings: no enlargement of the family pool can push a polynomial-identity system to full coverage of a square class, since the obstruction is Diophantine rather than computational. The greedy ceilings of Table 1 are therefore not a failure of the search but the expected signature of the barrier.

The theorem does not touch per-prime infinite constructions (Assumption 2), which remain the live angle. A further limitation of the present work: the greedy set-cover is heuristic, so the 73.5–84.9% figures are lower bounds on what finite systems achieve rather than exact maxima; and the 423 vacuous “prime-covering” progressions show the discovery filter needs a $\gcd(A, B) = 1$ tightening.

6 Conclusion

We have proved (Theorem 1) that no finite system of polynomial identities (1) can cover all primes of any square class modulo 840, via the Schinzel–Mordell non-residue obstruction and a CRT–Dirichlet construction. The result is supported by exact computation: Jacobi symbol -1 on all 5,965 admissible progressions of a 21,296-family pool, greedy coverage ceilings of 73.5–84.9%, and the explicit escaping prime 77,598,049.

7 References

References

- [1] A. Schinzel. [PLACEHOLDER: full journal reference for Schinzel’s 1956 nonexistence result on Erdős–Straus identities — unverified].
- [2] L. J. Mordell. [PLACEHOLDER: full journal reference for Mordell’s 1967 paper — unverified].
- [3] Wikipedia contributors. *Erdős–Straus conjecture*. [PLACEHOLDER: access date and revision unverified].
- [4] [PLACEHOLDER: citation for the Ionascu–Wilson per-prime construction idea mentioned in Assumption 2 — unverified].